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Prop ideal primo iff cociente di integridad

Proposición 1. *Sea $I \subsetneq R$ un ideal, entonces*

$$I \text{ es un ideal primo} \iff {}^{R}\!/_{I} \text{ es un dominio de integridad.}$$

***Demostración*:** Podemos ver la equivalencia directamente:

$$\begin{aligned} I \text{ es primo} &\iff I \neq R \wedge (\forall a, b \in R : ab \in I \implies a \in I \vee b \in I) \\ &\iff {}^{R}\!/_{I} \neq \{0\} \wedge \left(\forall \overline{a}, \overline{b} \in {}^{R}\!/_{I} : \overline{a}\overline{b} \overset{\substack{\text{absorción} \\ \downarrow}}{=} ab + I = \overline{0} \implies \overline{a} = \overline{0} \vee \overline{b} = \overline{0}\right) \\ &\iff {}^{R}\!/_{I} \text{ es un dominio de integridad.} \end{aligned}$$

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Referenciado en

- `prop-variedad-irreducible-iff-anillo-coordenadas-dominio-integridad`
- `teo-extension-entera-ideal-primo-maximal-iff-maximal`
- `cor-ideal-maximal-imp-primo`

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